

Shiqing Gao

Undergraduate Student in Mathematics

Hong Kong
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EDUCATION

The Chinese University of Hong Kong

B.Sc. in Mathematics, expected 2028

RESEARCH INTERESTS

Computational Inverse Problems Scientific Computing Numerical Methods Uncertainty Quantification
Machine Learning-related Inverse Problems Applied Mathematics

SELECTED PROJECTS

Image Inverse Problems: Denoising and Deblurring

- Completed a reproducible undergraduate computational study of denoising and deblurring under controlled synthetic degradations.
- Compared Gaussian filtering, Tikhonov regularization, TV denoising, non-local means, Wiener deconvolution, Richardson-Lucy deconvolution, and a small CNN baseline.
- Evaluated PSNR, SSIM, runtime, visual comparisons, error maps, parameter sensitivity, and degradation robustness.

Tools: Python, NumPy, pandas, matplotlib, scikit-image

Subdiffusion Mode Decay

- Completed a toy spectral visualization comparing Fourier mode decay in normal diffusion and time-fractional subdiffusion.
- Used a sine eigenbasis on $\Omega = (0, 1)$ with exponential and Mittag-Leffler-type decay factors.
- Documented the limitation that the project is a visualization, not a full inverse solver.

Tools: Python, NumPy, matplotlib

Machine Learning Uncertainty Quantification

- Completed a reproducible regression uncertainty project using the UCI Energy Efficiency dataset with Heating Load as the target.
- Compared approximate Gaussian process intervals, split conformal prediction, naive bootstrap intervals, and residual-corrected bootstrap intervals.
- Studied empirical coverage, interval width, repeated-split stability, reduced-training-size behavior, and perturbed-input robustness.

Tools: Python, NumPy, pandas, scikit-learn, matplotlib

Mathematical Modeling and Optimization

- Completed a chip thermal management modeling workflow using surrogate models for thermal resistance, pressure drop, and temperature nonuniformity.
- Used Pareto filtering, weighted-sum ranking, TOPSIS, weight-scenario analysis, random weight sampling, and Monte Carlo perturbation checks.
- Produced tables, figures, and written analysis focused on assumptions, trade-offs, robustness, and limitations.

Tools: Python, NumPy, pandas, matplotlib, LaTeX

TECHNICAL SKILLS

Programming: Python, C++, TypeScript basics

Scientific Computing / Data: NumPy, pandas, matplotlib, scikit-image, scikit-learn

Web: Astro, HTML, CSS, Vercel

Writing / Tools: LaTeX, Markdown, Git

COURSEWORK

Linear Algebra I Honours Linear Algebra II Advanced Calculus I-II Mathematical Analysis I
Complex Variables with Applications Basic Concepts in Statistics and Probability I
Fundamental Computing with C++ Data Structures and Applications